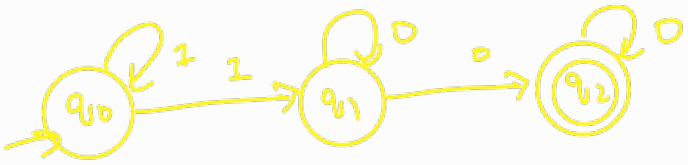
(NFA) → not all states finite



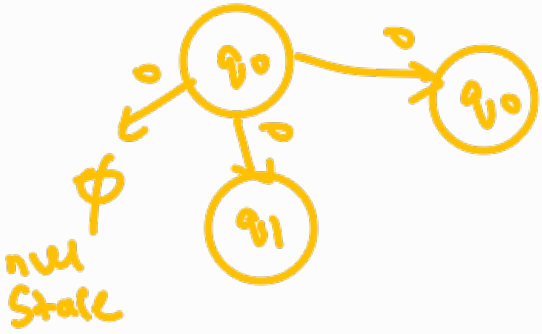
$(Q, \Sigma, q_0, F, \delta)$

$\delta : Q \times \Sigma \to 2^Q$



After zero comes as input I have only one choice that is to go to q_1.

→ DFA



Multiple curves to go to any state → NFA

Simple string $|w| = 1$ a, b, c

Simple Language (made from Simple string) $\{a, b, c\}$

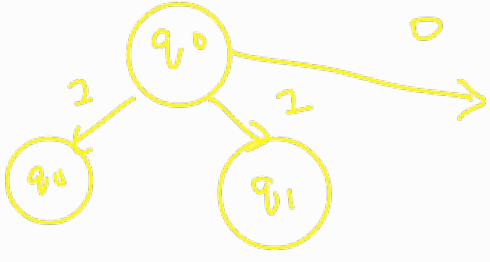
Regular Language → All Simple Languages are Regular.

↳ we can make its R.E

↓

DFA / NFA exist

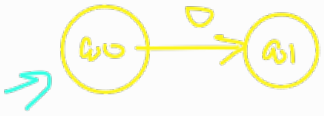




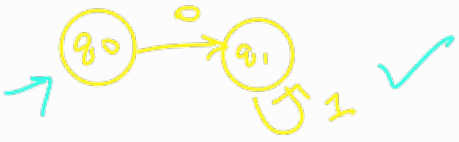
multiple choices

Note: In dfa we have to define all states (including the trap) with inputs.

$\Sigma = \{0, 1\}$

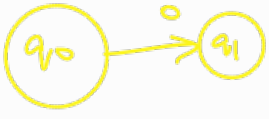


(X) we also have to define for 1.

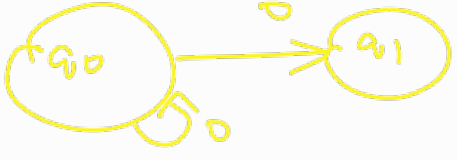


✓

But in NFA we do this: Define for one input only.



In DFA for each state and input symbol there can be only one next state.



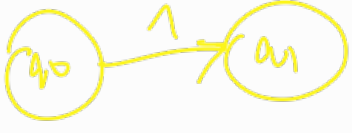
(X)

here at 0 we are going to both q_0, q_1

In NFA null move is allowed / meaning without consuming any input we move to next state.)

↓

not allowed in DFA



NFA → DFA

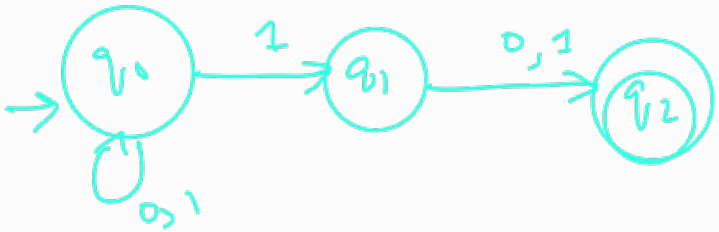
n(states) → max possible states (2^n)

Q NFA of all Binary strings in which 2nd last Symbol is 1.

$\Sigma(0,1)$ = 1, 0, 1, 00, 01, 11, 10, 000, 010, 110, 100, 111, 001

Acceptable 010, 110, 111, 10, 11 → min

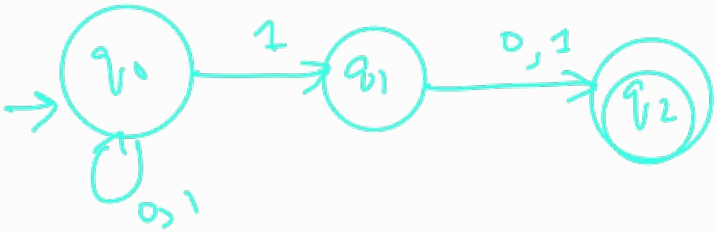
Anything 1 → Anything



no need of trap in NFA.

Converting from NFA to DFA

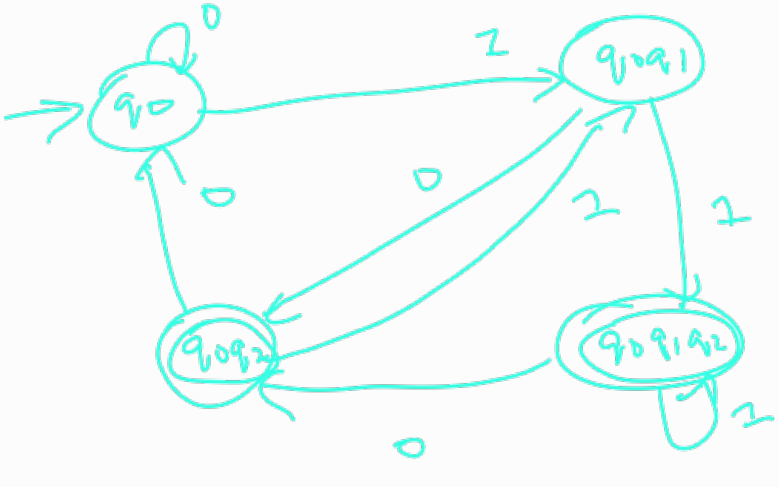
1) Make State diagram.



States \ Input	0	1
→ q_0	q_0	q_1 q_0
q_1	q_2	q_2
q_2	ϕ	ϕ

Now making DFA State table.

States	0	1
→ q_0	q_0	q_0q_1
q_0q_1	q_0q_2	q_0q_1 q_2
q_0q_2	q_0 ϕ	q_0q_1 ϕ

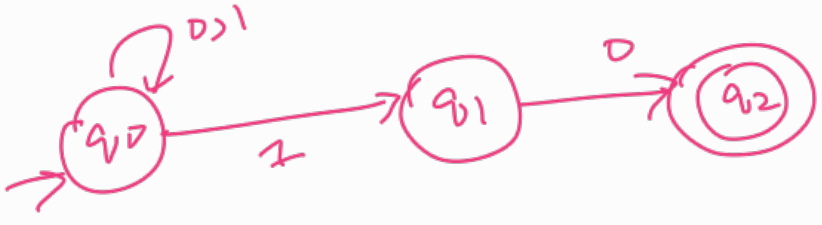


DFA

q_0 q_1 q_2	$q_0q_1\phi$	$q_0q_1q_2\phi$

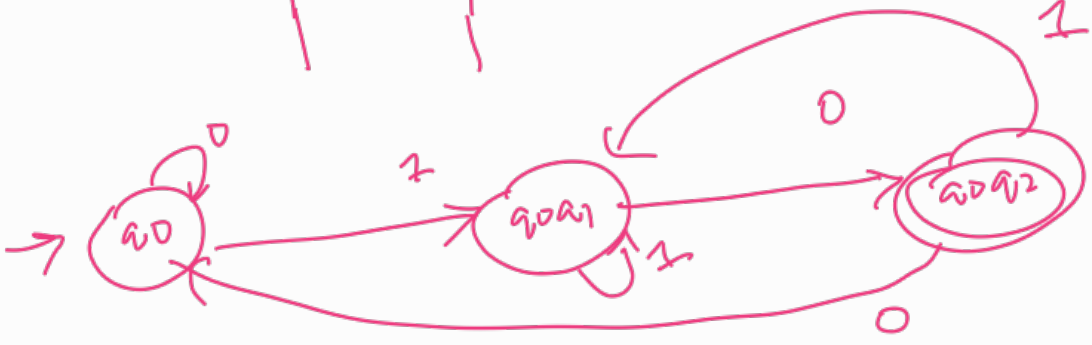
In NFA q_2 was Final so in dFA every state that contains q_2 will be final.

NFA



	0	1
→ q_0	q_0	q_1q_0
q_1	q_2	ϕ
q_2	ϕ	ϕ

States	0	1
q_0	q_0	q_0q_1
q_0q_1	q_0q_2	q_1q_0
q_0q_2	q_0	q_0q_1



(NULL - NFA) generally input Symbols are 0 and 1.

But now we are using null move. Now Input Symbol is also $\wedge$.



In this will 'a' be accepted

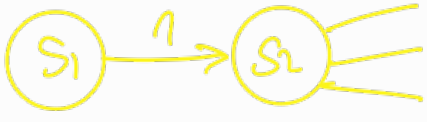
Ans: yes $a \wedge \wedge \rightarrow a$

$\wedge b \wedge \rightarrow b$

$\wedge \wedge c \rightarrow c$

---

converting (NULL - NFA to NFA) $\Rightarrow$ eliminating null-moves



1) find all edges starting from S_2
2) Duplicate all edges to S_1 without changing the edge labels
3) if S_1 is initial make S_2 initial
4) if S_2 is final make S_2 final



every state on ^ goes to itself. *

^-nfa → nfa



States \ Input	a	b	c	^
q_0	q_0	ϕ	ϕ	q_1
q_1	ϕ	q_1	ϕ	q_2
q_2	ϕ	ϕ	q_2	ϕ

$q_0 : q_1\ q_1\ q_2$

$q_1 : q_1\ q_2$

$q_2 : q_2$

} null-closure

States we can reach by null including the state itself

NFA Table

	a	b	c
→ q_0	$q_0 q_1 q_2$	q_1, q_2	q_L
* q_1	ϕ	$\{q_1, q_3\}$	$\{q_2\}$
* q_2	ϕ	ϕ	$\{q_3\}$

if the null closure has the final state then more than one state as final

$(q_0, a) \rightarrow q_0 \cup \phi \cup \phi \rightarrow q_0$

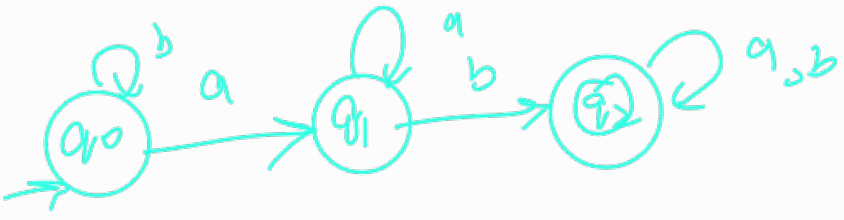
$= (q_0, q_1, q_2)$

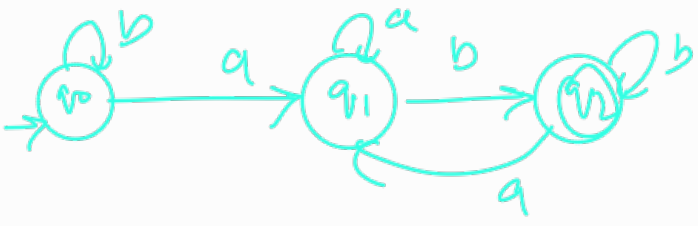
DFA practice:

$\Sigma = \{a,b\}^* \rightarrow \lambda, a, b, aa, ab, ba, bb, aaa, aba, baa, bba, aab, abb, bab, bbb.$

waw
wabbw

ab —
aba —
baababbb —
aabaabbb —
baaab —





Accepts string that ends with ab.

min ab, aab..

abbbab
aaabaaab

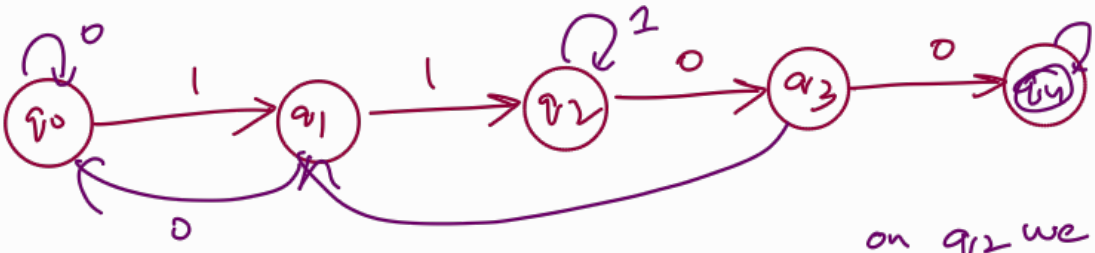
Design dfa that accepts all string containing 1100 as substring

$L = \Sigma^* \{1100\} = \lambda, 0, 1, 00, 01, 10, 11, \ldots$

min → 1100

w1100

01100
1100
111100
1111100
0101100



on q2 we already applied 0 now we need 1

1

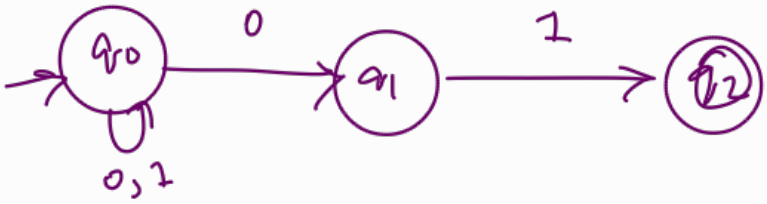
0111100

NFA - problems: $\Sigma = \{0, 1\}$ 0, 1, 00, 01, 10, 11, 000, 010, 100, 110, 001, 011, 101, 111 ..

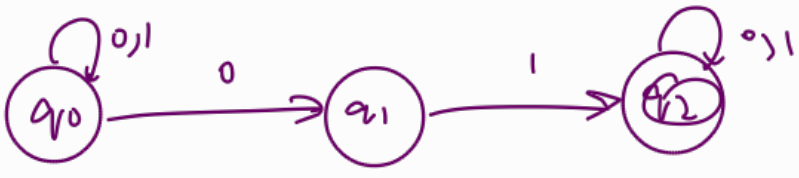
Starts with 01



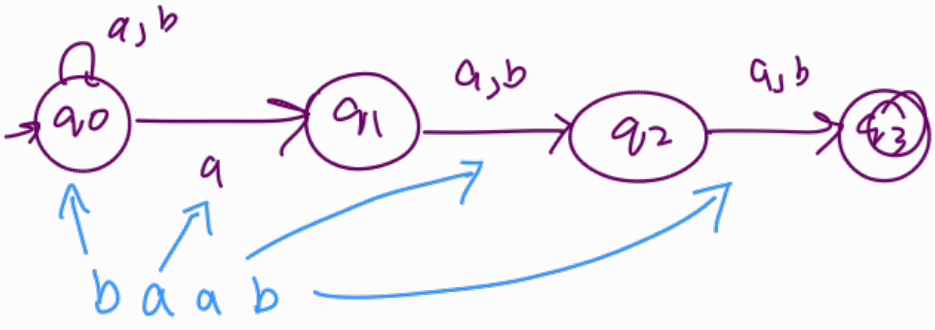
ends with 01



Substring 01



Third Symbol from the Right end is (a).



a a b b ←
↑
b b a a b ←

(Regular expression) → to represent regular language.

$R_1 + R_2$ → or

$R_1 \cdot R_2$ → concatenate (followed by.

Any no of a's followed by any no of b's followed by any number of c's.

$a^* b^* c^* \rightarrow a^+ b^+ c^+$ λ is not possible

even length string over $\Sigma = \{0\}$

$\{\lambda, 00, 0000, 0^6\}$

0 2 4 6

$(00)^*$

Odd length strings over $\Sigma = \{1\}$ $\{1, 111, 11111 \ldots$

$(11)^* 1$ at 1 → 1

at 1 → 111

at 2 → 11111

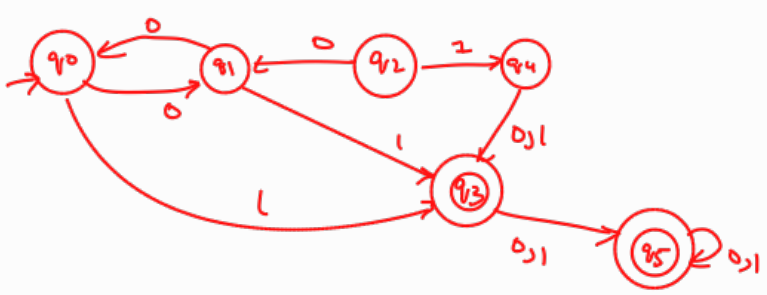
no of a's divisble by 3. $= \{\lambda, a^3, a^6 \ldots\}$

$(b^* a b^* a b^* a b)^*$

→ $(b^* b^* b^* b^*)^*$ zero no of a's.

for λ.

min of dfa (Reducing the no of states in dfa)



State	0	1
q_0	q_1	q_3
q_1	q_0	q_3
q_2	q_1	q_4
q_3	q_5	q_5
q_4	q_3	q_3
q_5	q_5	q_5

1) Remove all unreachable states.

Deadlock States

↳ States that we can reach from initial State

q_2 and q_4 unreachable

eliminate them

State	0	1
→ q_0	q_1	q_3
q_1	q_0	q_3
~~q_2~~	~~q_1~~	~~q_4~~
q_3^+	q_5	q_5
~~q_4~~	~~q_3~~	~~q_3~~
q_5^+	q_5	q_5

[non final] [final]

[q_0 q_1] [q_3 q_5]

Checking both q_0, q_1 ~~for~~ 0 and 1 and see if it still remains in the same set.

$q_0, q_1 \xrightarrow{0} q_1, q_0$

$q_0, q_1 \xrightarrow{1} q_3, q_3$

(X)

doesn't belong in the same set of [q_0, q_1]

[q_0, q_1] [q_3 q_5]

e.g

	0	1
→A	B	C
B	B	D
C	B	C
D	B	E
(E)	B	C

{A, B, C, D} {E}

Checking A, B

Both Belong to same set after applying 0, 1.

Now checking C, you can check C with either A and B

lets do A, C

{A, B, C} {D} {E}

{A, B, C} {D} {E}

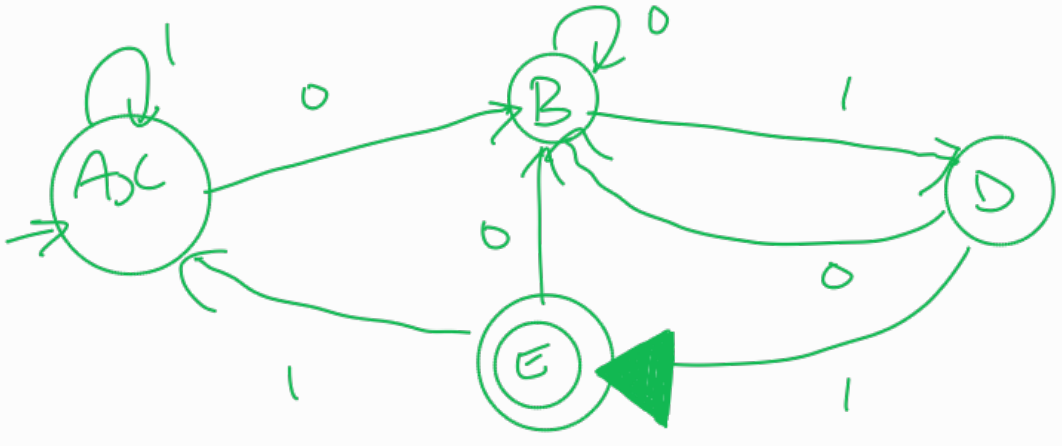
now checking A, B, C

{A, C} {B} {D} {F}

also checking B and C

Now checking only {A, C} as the rest three are seperate.

{A, C} {B} {D} {F}



now Split the transion table into two tables

T1 and T2:
- T1 → contains all final states
- T2 → contains non-final states.

DFA-min

Draw a transion table, Remove all the unreachable states.

	0	1
→ q_0	q_1	q_5
q_1	q_6	q_2
* q_2	q_0	q_2
q_3	q_7	q_5
q_4	q_2	q_6
q_5	q_6	q_4
q_6	q_6	q_2
q_7	q_6	q_2

non-f f

$[q_0\ q_1\ q_4\ q_5\ q_6\ q_7)\ [q_2]$

1-eq

$\delta(q_0,0) = q_1 \quad \delta(q_1,0) = q_5$

$\delta(q_0,1) = q_6 \quad \delta(q_1,1) = q_2$ (X)

So $q_0 \neq q_1$

$\delta(q_0,0) = q_1 \quad \delta(q_4,0) = q_5$

$\delta(q_0,1) = q_6 \quad \delta(q_4,1) = q_5$

$q_0 = q_4$

$\delta(q_0,0) = q_1 \quad \delta(q_5,0) = q_2$

$\delta(q_0,1) = q_6 \quad \delta(q_5,1) = q_6$ (X)

$q_0 \neq q_5$

now compare q_1 with q_5 to see if they are equal so we can place them in same set.

$\delta(q_1,0) = q_6 \quad \delta(q_1,1) = q_2$

$\delta(q_5,0) = q_2 \quad \delta(q_5,1) = q_6$ (X)

$q_1 \neq q_5$

$\delta(q_0, 0) = q_1$ $\delta(q_0, 1) = q_5$

$\delta(q_6, 0) = q_6$ $\delta(q_6, 1) = q_4$

$q_0 = q_6$

Similarly

$q_0 \neq q_2$

Compare q_7 with q_1 → $q_1 = q_7$

$\{q_0, q_4, q_6\}$ $\{q_1, q_7\}$ $\{q_2\}$ $\{q_5\}$

2-eq

$\delta(q_0, 0) = q_1$ $\delta(q_0, 1) = q_5$

$\delta(q_4, 0) = q_7$ $\delta(q_4, 1) = q_5$

$q_0 = q_4$

$q_0 \neq q_6$

$\{q_0\ q_4\}\ \{q_6\}\ \{q_1, q_7\}\ \{q_2\}$

$\{q_5\}$

$q_1 = q_7$

final as they are present in same set.

$[q_0\ q_4]$ $[q_6]$ $[q_1\ q_7]$ $[q_2]$ $[q_5]$

cyk algo. checking whether a string is valid in CFG.

baaba → len = 5
12345

S → AB | BC
A → BA | a
B → CC | b
C → AB | a

	5	4	3	2	1
1	(1,5)	(1,4) ∅	(1,3) ∅	(1,2) A,S	(1,1) B
2	(2,5)	(2,4) B	(2,3) B	(2,2) A,C	
3	(3,5) B	(3,4) S,C	(3,3) A,C		
4	(4,5) A,S	(4,4) B			
5	(5,5) A,C				

(Rightmost x first down)

top to down diagonal fill

first fill
(1,1) → b
(2,2) → a
(3,3) → a
(4,4) → b
(5,5) → a

(1,1) · (2,3)
(1,2) x (3,3)

Since b is generated by which prod B → CC | b ✓

(1,2) → (1,1) · (1,2)
B · A,C
(BA, BC)
(A, S)

(2,3) → (2,2) · (3,3)
(A,C) (A,C)
AA, AC CA, CC
(X) (X) (X) B

(1,3) → (1,1) x (2,3) 4
(1,2) x (3,3)

(1,4) → (1,1) x (2,4) 4
(1,2) x (3,4) 4
(1,3) x (4,4)

→ B · B

→ (A,S) (S,C)
AS, AC, SS, SC
x x x x

→ ϕ · (4,4) → ϕ